

Carlos Bilbao Lara



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AI & Software Developer and Engineer with over 3 years of experience, involved in international projects for top-tier clients such as HPE, AT&T, Telefónica and Verisure. Currently focused on leading machine learning and Artificial Intelligence projects, having developed innovative solutions and applications that automate processes through AI, achieving 30%-time savings for users.

[GitHub](#), [LinkedIn](#), [Portfolio](#)

PROFESIONAL EXPERIENCE

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Madrid, Spain

AI & Software Lead

May 2025 – now

- Led enterprise AI adoption across client development teams, achieving a **+600% increase in AI usage within one year**, while presenting progress, adoption metrics and strategic recommendations directly to the **client's Director** through technical reports and executive updates.
- Led the **design, develop and govern MCP** servers using Python, Java and TypeScript, enabling AI assistants and agents to securely interact with enterprise tools, APIs and knowledge sources.
- Designed and implemented **multi-agent systems with LangGraph**, including agent orchestration, tool calling, state management and automated workflows, translating client needs into scalable AI Proofs of Concept
- **Created Power BI dashboards and data pipelines** to monitor AI adoption, usage, productivity and maturity metrics.
- Developed supporting backend and data infrastructure using Python, Java microservices, MongoDB, SQL Server, Helm, **Kubernetes**, Jenkins, GitHub Actions, SonarQube and CI/CD practices....

Full-On-Net

Madrid, Spain

Artificial Intelligence Engineer

June 2024 – May 2025

- Engineered Artificial Intelligence solutions utilizing **NLP, Vision Computing**, and multimodal LLMs.
- Developed AI pipelines incorporating microservices, leveraging **Hugging Face, PyTorch, Flask, NumPy**, OpenCV, **Pandas**, and SQL Alchemy.
- **Led** the development of microservices architecture, improving scalability and maintainability.
- Conducted research and implemented **cutting-edge models**.
- Designed an innovative generic class, decreasing the time needed to modify and test new models by **30%**.
- Architected a microservices framework using **Azure**.
- **Client collaboration**: Worked closely with clients to understand their needs and key objectives, actively participating in meetings and discussions to define tailored solutions.
- Joint problem-solving: Facilitated decision-making through a collaborative approach, reaching agreements on solutions that maximize value for both parties.
- **Client relationship management**: Acted as a liaison between the technical team and the client, ensuring clear expectations and delivering satisfactory results through effective and ongoing communication.
- Collaborated with my team utilizing Git, GitHub, and Agile methodologies.
- Deployed software using Ansible, Docker, and **Kubernetes**.

CDS an HPE Company

Software/DevOps Engineer

Madrid, Spain

January 2023–June 2024

- Developed applications for the **R&D** department of CTG, with an emphasis on Continuous Development (CD) and Continuous Integration (CI).
- Conducted backend development using Java 11 and Python 3, incorporating libraries such as NumPy, pandas, PyTorch, and Hugging Face.
- Initiated the development of NLP transformers, data analysis, and data preparation to enhance the product.
- Oversaw the migration from Java 8 to Java 11, **Python** 3.6 to 3.11, and AngularJS to Angular.
- Engineered complex functions and restructured code, achieving a **40%** improvement in efficiency.
- Implemented **CI/CD** pipelines using Jenkins, encompassing the creation of the pipeline with testing, SonarQube analysis, build, and deployment phases.

Inetum

Software Developer

Madrid, Spain

May 2021–December 2021

- Designed and implemented architecture from scratch.
- Developed web applications using React and Java.

EDUCATION

Universidad Politécnica de Madrid

Computer Science, bachelor's degree

Final Project: CNN able to recognize emoticons in any picture.

Madrid, Spain

September 2018-June 2022

Universidad Alfonso X

Artificial Intelligence, Master

Final Project: Designed and implemented a **hybrid neuroevolution algorithm**

Madrid, Spain

September 2023- Sept 2024

Universidad Politécnica de Madrid

Artificial Intelligence, PhD

Working on: **Hybrid Neuroevolution**

Madrid, Spain

September 2025

OTHER SKILLS

- Native Spanish speaker, fluent in **English (C1 level)**, with all professional work conducted primarily in English.
- Proficient in both relational and non-relational databases.
- Skilled in data analysis.
- Strong problem-solving abilities.
- Demonstrated commitment to work.
- **Knowledgeable in best practices.**
- Tennis and padel coach and player.
- Communicative and always willing to help others.
- Time management.